STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 7

Description

7. Demonstration of clustering rule process on dataset student.arff using hierarchical clustering.

Aim: This experiment illustrates the use of hierarchical clustering with Weka explorer. The sample data set used for this example is based on the student data available in ARFF format. This document assumes that appropriate preprocessing has been performed. This istudent dataset includes 14 instances.
Steps involved in this Experiment
Step 1: Run the Weka explorer and load the data file student.arff in preprocessing interface.
Step 2: Inorder to perform clustering select the 'cluster' tab in the explorer and click on the choose button. This step results in a dropdown list of available clustering algorithms.
Step 3 : In this case we select 'hierarchical'.
Step 4: Next click in text button to the right of the choose button to get popup window shown in the screenshots. In this window we enter six on the number of clusters and we leave the value of the seed on as it is. The seed value is used in generating a random number which is used for making the internal assignments of instances of clusters.
Step 5 : Once of the option have been specified. We run the clustering algorithm there we must make sure that they are in the 'cluster mode' panel. The use of training set option is selected and then we click 'start' button. This process and resulting window are shown in the following screenshots.
Step 6 : The result window shows the centroid of each cluster as well as statistics on the number and the percent of instances assigned to different clusters. Here clusters centroid are means vectors for each clusters. This clusters can be used to characterized the cluster.
Step 7: Another way of understanding characteristics of each cluster through visualization ,we can do this, try right clicking the result set on the result. List panel and selecting the visualize cluster assignments. Interpretation of the above visualization From the above visualization, we can understand the distribution of age and instance number in each cluster. For instance, for each cluster is dominated by age. In this case by changing the color dimension to other attributes we can see their distribution with in each of the cluster.
Step 8: We can assure that resulting dataset which included each instance along with its assign cluster. To do so we click the save button in the visualization window and save the result student hierarchical .The top portion of this file is shown in the following figure.

Description:
Hierarchical clustering involves creating clusters that have a predetermined ordering from top to bottom. For example, all files and folders on the hard disk are organized in a hierarchy. There are two types of hierarchical clustering, Divisive and Agglomerative.

Procedure:

Step 1: Load appropriate dataset into WEKA

Step 2: Go to Cluster tab->choose->Hierarchical clustering

Procedure and Overvation:

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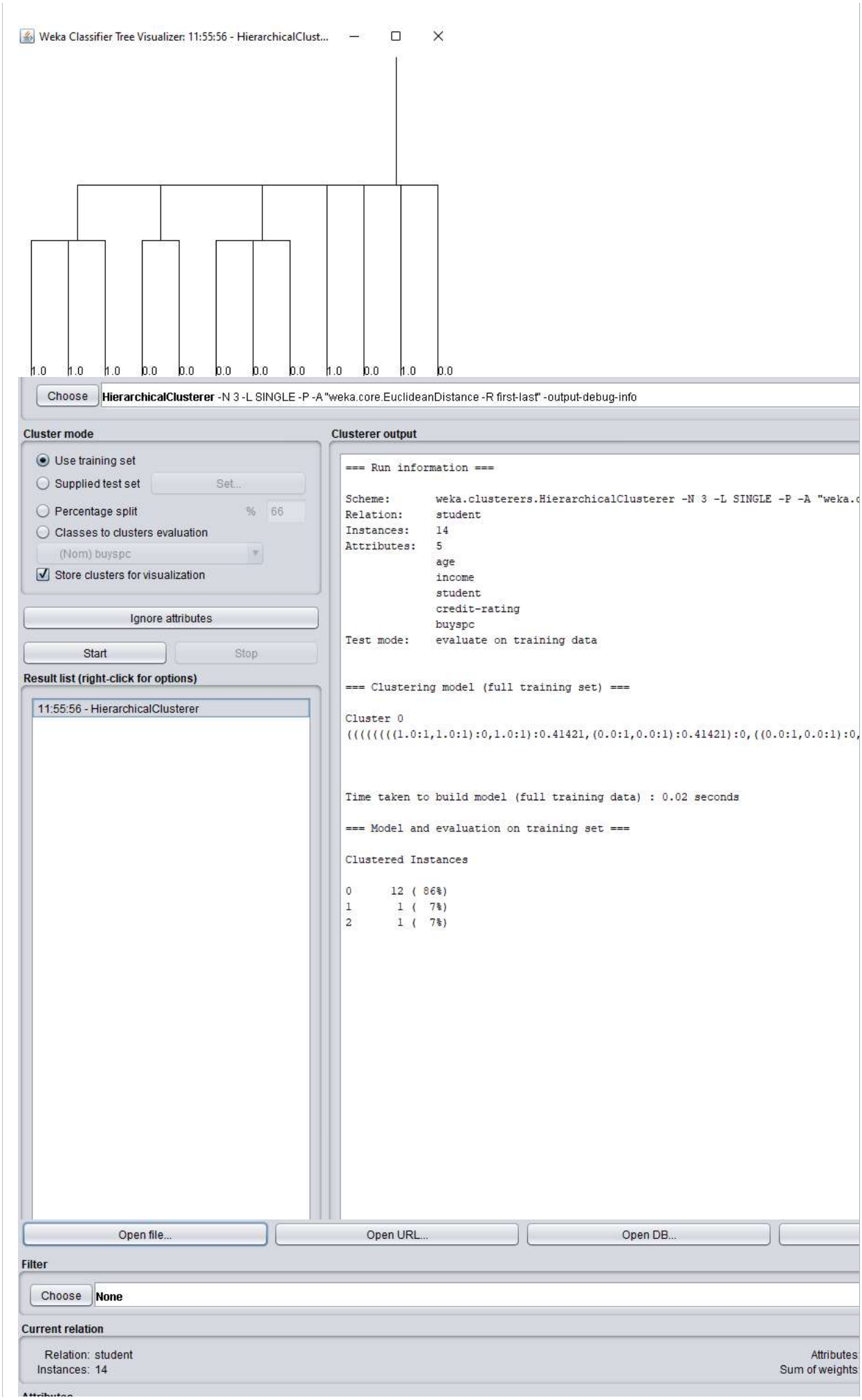
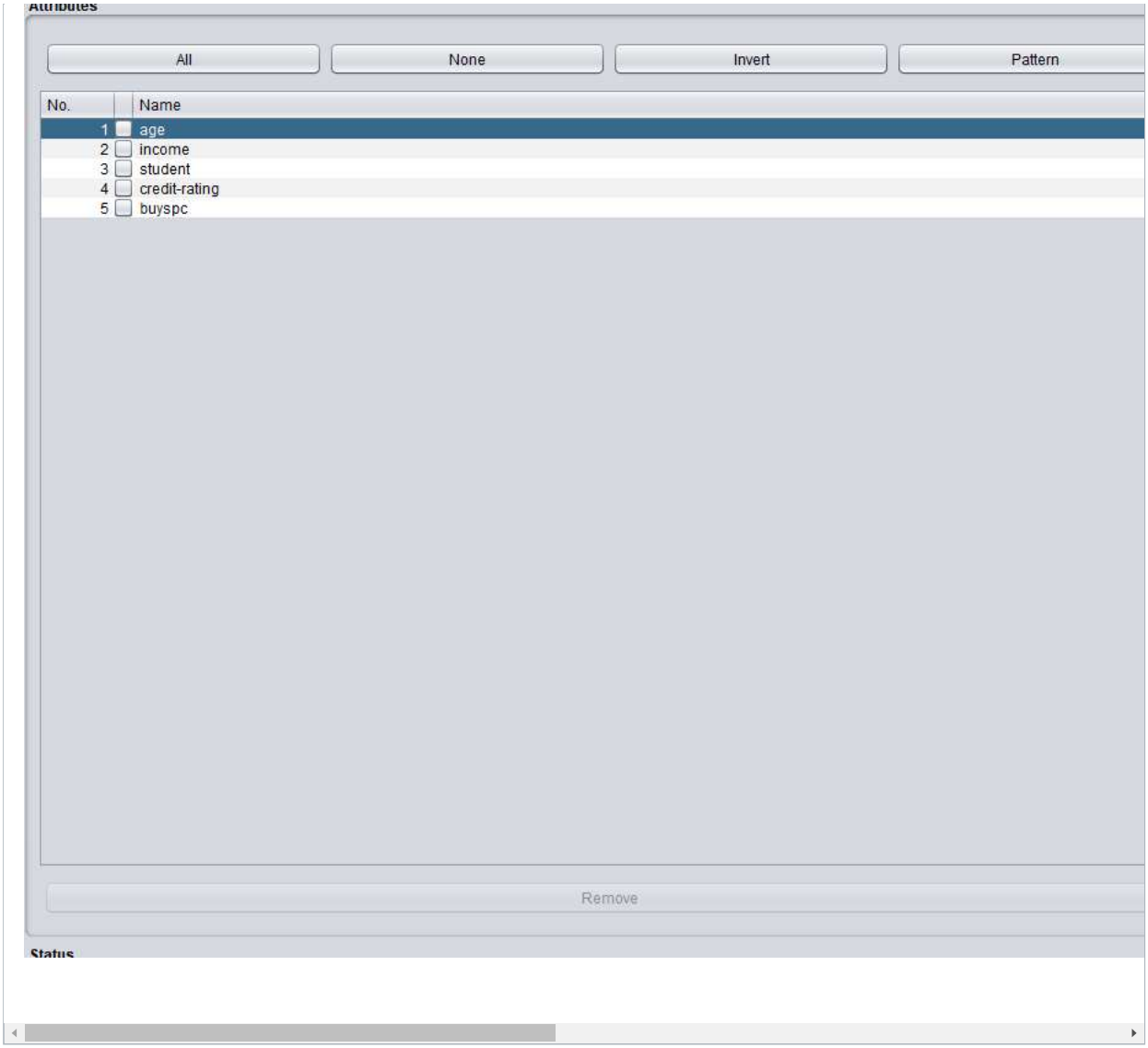
Step 1: Load appropriate dataset into WEKA

Step 2: Go to Cluster tab→choose→Hierarchical clustering

Dataset student .arff

```
@relation student

@attribute age {<30,30-40,>40}

@attribute income {low, medium, high}

@attribute student {yes, no}

@attribute credit-rating {fair, excellent}

@attribute buyspc {yes, no}

@data

%

<30, high, no, fair, no

<30, high, no, excellent, no

30-40, high, no, fair, yes

>40, medium, no, fair, yes

>40, low, yes, fair, yes

>40, low, yes, excellent, no

30-40, low, yes, excellent, yes

<30, medium, no, fair, no

<30, low, yes, fair, no

>40, medium, yes, fair, yes

<30, medium, yes, excellent, yes

30-40, medium, no, excellent, yes

30-40, high, yes, fair, yes

>40, medium, no, excellent, no

%
```


Weka Classifier Tree Visualizer: 11:55:56 - HierarchicalClust...
1.0 1.0 1.0 0.0 0.0 0.0 0.0 0.0 1.0 0.0 1.0 0.0
Choose
HierarchicalClusterer -N 3 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last" -output-debug-info
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) buyspc
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
11:55:56 - HierarchicalClusterer
Clusterer output
=== Run information ===
Scheme: weka.clusterers.HierarchicalClusterer -N 3 -L SINGLE -P -A "weka.c
Relation: student
Instances: 14
Attributes: 5
age
income
student
credit-rating
buyspc
Test mode: evaluate on training data
=== Clustering model (full training set) ===
Cluster 0
((((((((1.0:1,1.0:1):0,1.0:1):0.41421,(0.0:1,0.0:1):0.41421):0,((0.0:1,0.0:1):0,
Time taken to build model (full training data) : 0.02 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 12 (86%)
1 1 (7%)
2 1 (7%)
Open file...
Open URL...
Open DB...
Filter
Choose
None
Current relation
Relation: student
Instances: 14
Attributes
Sum of weights



VIVA RESPONSES

Q) _____ is data about data

Metadata

Q) How many categories of functions are involved in data mining?

4

Q) How many approaches are there in data warehousing to integrate heterogeneous databases?

2

Q) What does data mining system classification consist of?

All of the above

Q) What does DMQL stand for?

Data mining query language